Sarah Thomas

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Education

Vellore Institute of Technology September 2023 – May 2027
B.Tech in Computer Science – Specialization in Cloud Computing and Automation *CGPA: 8.81/10*

St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh

- All India Senior School Certificate Examination (March 2022 – May 2023) *CGPA: 7.88/10*
- All India Secondary School Examination (March 2020 – June 2021) *CGPA: 9.2/10*

Experience

MPOnline Limited (Govt. of Madhya Pradesh & TCS Joint Venture) 20 May 2026 – 31 Aug 2026
Software Engineering & AI Foundation Intern *Remote*

- Collaborated within a cross-functional, 7-member team under Agile workflows to architect an AI-Powered Smart Logistics System using ASP.NET Core, EF Core, and SignalR for real-time tracking.
- Engineered automated backend workflows and modern digital infrastructure by integrating generative AI frameworks, Retrieval-Augmented Generation (RAG) models, and n8n orchestration automation.
- Applied advanced object-oriented programming paradigms to design robust, production-ready database architectures while executing rigorous unit and integration testing.

Technical Skills

Cloud & DevOps: AWS (Lambda, S3, API Gateway, EC2, IAM, CloudWatch), Docker, Docker Compose, Prometheus (Metrics Telemetry), Terraform, CI/CD (GitHub Actions), Git, GitHub, Boto3 (AWS SDK)

Languages & Frameworks: Java, Python, C#, Linux/Bash, FastAPI, ASP.NET Core, REST APIs

Databases & Tools: MongoDB, Microsoft SQL Server, EF Core, OpenAI API, n8n, SignalR, Pandas, Matplotlib

Projects

AI Incident & Log Analysis Assistant github.com/THOMAS-SARAH/exception_analyzer
Tools: Python, FastAPI, OpenAI API

- Architected a RESTful AI incident response microservice utilizing FastAPI and OpenAI to process server log traces and automate Root-Cause Analysis (RCA) report generation.
- Formulated structured prompt engineering workflows to analyze complex system telemetry errors, delivering actionable remediation steps for technical operations teams.

Serverless Multi-Tenant Configuration Processor github.com/THOMAS-SARAH/aws-serverless-config-processor
Tools: AWS Lambda, S3, API Gateway, Python (Boto3), IAM

- Built an event-driven serverless microservice using AWS Lambda and Python (Boto3) to process and safely persist multi-tenant SaaS configuration files into AWS S3 environments.
- Integrated AWS API Gateway HTTP endpoints with custom execution IAM roles to manage stateless event routing and bucket write operations securely.
- Configured dynamic storage partitioning and structured JSON response formatting to achieve fully automated and isolated tenant data management.

AI-Driven Cloud Resource & Cost Optimizer github.com/THOMAS-SARAH/multi-region-aws-telemetry-pipeline
Tools: Prometheus, Node Exporter, Terraform, AWS, Linux/Bash, FastAPI

- Deployed a real-time infrastructure telemetry pipeline using Prometheus and Node Exporter to monitor system performance, CPU/memory utilization, and cloud workload health.
- Provisioned multi-region cloud infrastructure via modular Terraform templates, integrating FastAPI backend endpoints for automated resource optimization and anomaly detection.

Certifications and Extra-Curricular

Certifications: ServiceNow University Virtual Internship Program; NPTEL – Internet Of Things, Cloud Computing and Distributed Systems; AWS Educate – Introduction to Generative AI.

Extra-Curriculars: Education Minister in School Council (Leadership & Governance); Best Girl Student of Batch 2022-23; Club Coordinator of E-Green Club (Project & Stakeholder Management).